

# Faculty Skill Development Portal

## Comprehensive Project Report, API Documentation, and Viva Preparation Guide

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### Executive Summary

This project delivers a full-stack Faculty Skill Development Portal that enables administrators to manage learning assets and monitor outcomes while faculty users plan skills, take assessments, enroll in programs/courses, and track growth progress.

- Backend: FastAPI with role-based authorization, JWT tokens, and SQLAlchemy data layer.
- Frontend: React + TypeScript with dashboard workflows for Admin and Faculty roles.
- Database: PostgreSQL schema managed using Alembic migrations.
- API Scope: 105 REST operations across 20 route modules.

### Project Objectives

- Digitize and centralize faculty upskilling workflows.
- Provide role-specific experiences (Admin vs Faculty).
- Integrate AI features for question generation and coaching.
- Enable measurable progress through analytics and structured learning paths.

### Technology Stack

| Layer    | Technologies                                             |
|----------|----------------------------------------------------------|
| Frontend | React 18, TypeScript, Vite, Tailwind CSS, TanStack Query |
| Backend  | FastAPI, Pydantic, SQLAlchemy, Alembic, JWT Auth         |
| Database | PostgreSQL                                               |
| Tooling  | Docker Compose, npm, Python virtual environment          |

### High-Level Architecture

- Client UI issues authenticated REST calls through API wrappers.

- FastAPI routers route requests to service-layer business logic.
- Services use SQLAlchemy models for persistence and query operations.
- RBAC checks are enforced via dependencies and token-derived role context.
- AI integrations support coaching and draft question generation features.

## **Major Functional Modules**

- Authentication and session identity (`auth`).
- Faculty profile, skill tracking, and personalized news (`faculty`).
- Programs, enrollments, tests, attempts, and practice sets.
- Growth plans and roadmaps for structured learning progression.
- Courses with modules, quizzes, assessments, and completion tracking.
- Discussion/query management and analytics reporting for admins.

# API Documentation

Swagger is available at `http://localhost:8000/docs` when the backend runs. The table below is generated from route source files and can be imported into Postman using the attached collection JSON.

| Module (Tag)   | Operation Count |
|----------------|-----------------|
| ai-coach       | 1               |
| ai-questions   | 7               |
| analytics      | 3               |
| attempts       | 7               |
| auth           | 2               |
| courses        | 21              |
| discussions    | 4               |
| enrollments    | 2               |
| faculty        | 13              |
| growth-plans   | 6               |
| health         | 1               |
| news           | 2               |
| practice-sets  | 4               |
| programs       | 5               |
| queries        | 4               |
| question-packs | 9               |
| roadmaps       | 4               |
| skills         | 2               |
| tests          | 5               |
| users          | 3               |

## Authentication Flow (Viva Ready)

- `POST /api/v1/auth/login` validates credentials and issues JWT access/refresh tokens.
- Client stores token and sends `Authorization: Bearer ` for protected endpoints.
- Protected dependencies read claims and enforce role checks for ADMIN/FACULTY operations.
- `GET /api/v1/auth/me` returns logged-in user profile for session bootstrap.

## **Viva Explanation Script**

Suggested 6-8 minute sequence:

- State problem and target users: faculty and institute administrators.
- Explain architecture: React frontend, FastAPI backend, PostgreSQL persistence.
- Show RBAC and JWT flow, then demonstrate protected API call behavior.
- Walk through one core journey: faculty login -> enroll -> complete assessment -> view analytics.
- Present AI-enhanced modules (AI coach and AI question generation) as productivity features.
- Close with testing, scalability, and future enhancements.

## **Common Viva Questions with Answers**

### **Q: Why FastAPI?**

A: FastAPI gives fast async performance, native OpenAPI docs, and strict Pydantic validation for reliable APIs.

### **Q: How is security handled?**

A: Password hashing plus JWT-based auth and role-checked dependencies control access.

### **Q: How do you maintain schema changes?**

A: Alembic migration scripts version-control database changes.

### **Q: How can this scale?**

A: Stateless APIs can scale horizontally behind a load balancer; PostgreSQL can be optimized with indexing and read replicas.

### **Q: What is the project impact?**

A: It centralizes faculty development tracking and gives measurable skill progression through structured workflows.

# Appendix A: Endpoint Inventory

Generated from `server/app/api/v1/routes` decorators.

## ai-coach

| Method | Path                  | Source                               |
|--------|-----------------------|--------------------------------------|
| POST   | /api/v1/ai-coach/chat | server/app/api/v1/routes/ai_coach.py |

## ai-questions

| Method | Path                                                      | Source                                   |
|--------|-----------------------------------------------------------|------------------------------------------|
| GET    | /api/v1/ai-questions/batches                              | server/app/api/v1/routes/ai_questions.py |
| GET    | /api/v1/ai-questions/batches/{batch_id}                   | server/app/api/v1/routes/ai_questions.py |
| POST   | /api/v1/ai-questions/batches/{batch_id}/approve/{index}   | server/app/api/v1/routes/ai_questions.py |
| POST   | /api/v1/ai-questions/batches/{batch_id}/publish           | server/app/api/v1/routes/ai_questions.py |
| PATCH  | /api/v1/ai-questions/batches/{batch_id}/questions/{index} | server/app/api/v1/routes/ai_questions.py |
| POST   | /api/v1/ai-questions/batches/{batch_id}/reject/{index}    | server/app/api/v1/routes/ai_questions.py |
| POST   | /api/v1/ai-questions/generate                             | server/app/api/v1/routes/ai_questions.py |

## analytics

| Method | Path                                   | Source                                |
|--------|----------------------------------------|---------------------------------------|
| GET    | /api/v1/analytics/department-summary   | server/app/api/v1/routes/analytics.py |
| GET    | /api/v1/analytics/faculty/{faculty_id} | server/app/api/v1/routes/analytics.py |
| GET    | /api/v1/analytics/me                   | server/app/api/v1/routes/analytics.py |

## attempts

| Method | Path                                  | Source                               |
|--------|---------------------------------------|--------------------------------------|
| POST   | /api/v1/attempts                      | server/app/api/v1/routes/attempts.py |
| GET    | /api/v1/attempts/faculty/{faculty_id} | server/app/api/v1/routes/attempts.py |
| GET    | /api/v1/attempts/me                   | server/app/api/v1/routes/attempts.py |
| GET    | /api/v1/attempts/{attempt_id}         | server/app/api/v1/routes/attempts.py |
| POST   | /api/v1/attempts/{attempt_id}/answers | server/app/api/v1/routes/attempts.py |
| POST   | /api/v1/attempts/{attempt_id}/finish  | server/app/api/v1/routes/attempts.py |
| POST   | /api/v1/attempts/{attempt_id}/submit  | server/app/api/v1/routes/attempts.py |

## auth

| Method | Path               | Source                           |
|--------|--------------------|----------------------------------|
| POST   | /api/v1/auth/login | server/app/api/v1/routes/auth.py |
| GET    | /api/v1/auth/me    | server/app/api/v1/routes/auth.py |

## courses

| Method | Path                                                           | Source                              |
|--------|----------------------------------------------------------------|-------------------------------------|
| GET    | /api/v1/courses                                                | server/app/api/v1/routes/courses.py |
| POST   | /api/v1/courses                                                | server/app/api/v1/routes/courses.py |
| GET    | /api/v1/courses/analytics                                      | server/app/api/v1/routes/courses.py |
| GET    | /api/v1/courses/my-enrollments                                 | server/app/api/v1/routes/courses.py |
| PUT    | /api/v1/courses/progress/{module_id}                           | server/app/api/v1/routes/courses.py |
| POST   | /api/v1/courses/progress/{module_id}/quiz                      | server/app/api/v1/routes/courses.py |
| DELETE | /api/v1/courses/{course_id}                                    | server/app/api/v1/routes/courses.py |
| GET    | /api/v1/courses/{course_id}                                    | server/app/api/v1/routes/courses.py |
| PUT    | /api/v1/courses/{course_id}                                    | server/app/api/v1/routes/courses.py |
| GET    | /api/v1/courses/{course_id}/assessment                         | server/app/api/v1/routes/courses.py |
| POST   | /api/v1/courses/{course_id}/assessment                         | server/app/api/v1/routes/courses.py |
| POST   | /api/v1/courses/{course_id}/assessment-questions               | server/app/api/v1/routes/courses.py |
| DELETE | /api/v1/courses/{course_id}/assessment-questions/{question_id} | server/app/api/v1/routes/courses.py |
| GET    | /api/v1/courses/{course_id}/attempt                            | server/app/api/v1/routes/courses.py |
| POST   | /api/v1/courses/{course_id}/enroll                             | server/app/api/v1/routes/courses.py |
| POST   | /api/v1/courses/{course_id}/modules                            | server/app/api/v1/routes/courses.py |
| DELETE | /api/v1/courses/{course_id}/modules/{module_id}                | server/app/api/v1/routes/courses.py |
| PUT    | /api/v1/courses/{course_id}/modules/{module_id}                | server/app/api/v1/routes/courses.py |
| POST   | /api/v1/courses/{course_id}/modules/{module_id}/quiz           | server/app/api/v1/routes/courses.py |
| DELETE | /api/v1/courses/{course_id}/modules/{module_id}/quiz/{quiz_id} | server/app/api/v1/routes/courses.py |
| GET    | /api/v1/courses/{course_id}/progress                           | server/app/api/v1/routes/courses.py |

## discussions

| Method | Path                                        | Source                                  |
|--------|---------------------------------------------|-----------------------------------------|
| GET    | /api/v1/discussions                         | server/app/api/v1/routes/discussions.py |
| POST   | /api/v1/discussions                         | server/app/api/v1/routes/discussions.py |
| GET    | /api/v1/discussions/{discussion_id}         | server/app/api/v1/routes/discussions.py |
| POST   | /api/v1/discussions/{discussion_id}/replies | server/app/api/v1/routes/discussions.py |

## enrollments

| Method | Path                   | Source                                  |
|--------|------------------------|-----------------------------------------|
| POST   | /api/v1/enrollments    | server/app/api/v1/routes/enrollments.py |
| GET    | /api/v1/enrollments/me | server/app/api/v1/routes/enrollments.py |

## faculty

| Method | Path                                                 | Source                              |
|--------|------------------------------------------------------|-------------------------------------|
| GET    | /api/v1/faculty                                      | server/app/api/v1/routes/faculty.py |
| GET    | /api/v1/faculty/me                                   | server/app/api/v1/routes/faculty.py |
| PATCH  | /api/v1/faculty/me                                   | server/app/api/v1/routes/faculty.py |
| GET    | /api/v1/faculty/me/news                              | server/app/api/v1/routes/faculty.py |
| GET    | /api/v1/faculty/me/news-preferences                  | server/app/api/v1/routes/faculty.py |
| PUT    | /api/v1/faculty/me/news-preferences                  | server/app/api/v1/routes/faculty.py |
| GET    | /api/v1/faculty/me/skill-suggestions                 | server/app/api/v1/routes/faculty.py |
| POST   | /api/v1/faculty/me/skills                            | server/app/api/v1/routes/faculty.py |
| DELETE | /api/v1/faculty/me/skills/{skill_id}                 | server/app/api/v1/routes/faculty.py |
| PATCH  | /api/v1/faculty/me/skills/{skill_id}                 | server/app/api/v1/routes/faculty.py |
| POST   | /api/v1/faculty/register-faculty                     | server/app/api/v1/routes/faculty.py |
| GET    | /api/v1/faculty/{faculty_id}                         | server/app/api/v1/routes/faculty.py |
| POST   | /api/v1/faculty/{faculty_id}/verify-skill/{skill_id} | server/app/api/v1/routes/faculty.py |

## growth-plans

| Method | Path                                          | Source                                   |
|--------|-----------------------------------------------|------------------------------------------|
| GET    | /api/v1/growth-plans                          | server/app/api/v1/routes/growth_plans.py |
| POST   | /api/v1/growth-plans                          | server/app/api/v1/routes/growth_plans.py |
| DELETE | /api/v1/growth-plans/me                       | server/app/api/v1/routes/growth_plans.py |
| GET    | /api/v1/growth-plans/me                       | server/app/api/v1/routes/growth_plans.py |
| PATCH  | /api/v1/growth-plans/tasks/{task_id}          | server/app/api/v1/routes/growth_plans.py |
| POST   | /api/v1/growth-plans/weeks/{week_id}/complete | server/app/api/v1/routes/growth_plans.py |

## health

| Method | Path           | Source                             |
|--------|----------------|------------------------------------|
| GET    | /api/v1/health | server/app/api/v1/routes/health.py |

## news

| Method | Path         | Source                           |
|--------|--------------|----------------------------------|
| GET    | /api/v1/news | server/app/api/v1/routes/news.py |

| Method | Path                | Source                           |
|--------|---------------------|----------------------------------|
| GET    | /api/v1/news/topics | server/app/api/v1/routes/news.py |

## practice-sets

| Method | Path                                  | Source                                    |
|--------|---------------------------------------|-------------------------------------------|
| POST   | /api/v1/practice-sets                 | server/app/api/v1/routes/practice_sets.py |
| GET    | /api/v1/practice-sets/me              | server/app/api/v1/routes/practice_sets.py |
| GET    | /api/v1/practice-sets/{set_id}        | server/app/api/v1/routes/practice_sets.py |
| POST   | /api/v1/practice-sets/{set_id}/submit | server/app/api/v1/routes/practice_sets.py |

## programs

| Method | Path                          | Source                               |
|--------|-------------------------------|--------------------------------------|
| GET    | /api/v1/programs              | server/app/api/v1/routes/programs.py |
| POST   | /api/v1/programs              | server/app/api/v1/routes/programs.py |
| DELETE | /api/v1/programs/{program_id} | server/app/api/v1/routes/programs.py |
| GET    | /api/v1/programs/{program_id} | server/app/api/v1/routes/programs.py |
| PATCH  | /api/v1/programs/{program_id} | server/app/api/v1/routes/programs.py |

## queries

| Method | Path                       | Source                              |
|--------|----------------------------|-------------------------------------|
| GET    | /api/v1/queries            | server/app/api/v1/routes/queries.py |
| POST   | /api/v1/queries            | server/app/api/v1/routes/queries.py |
| GET    | /api/v1/queries/mine       | server/app/api/v1/routes/queries.py |
| PATCH  | /api/v1/queries/{query_id} | server/app/api/v1/routes/queries.py |

## question-packs

| Method | Path                                           | Source                                     |
|--------|------------------------------------------------|--------------------------------------------|
| GET    | /api/v1/question-packs                         | server/app/api/v1/routes/question_packs.py |
| POST   | /api/v1/question-packs                         | server/app/api/v1/routes/question_packs.py |
| GET    | /api/v1/question-packs/questions               | server/app/api/v1/routes/question_packs.py |
| DELETE | /api/v1/question-packs/questions/{question_id} | server/app/api/v1/routes/question_packs.py |
| PATCH  | /api/v1/question-packs/questions/{question_id} | server/app/api/v1/routes/question_packs.py |
| DELETE | /api/v1/question-packs/{pack_id}               | server/app/api/v1/routes/question_packs.py |
| GET    | /api/v1/question-packs/{pack_id}               | server/app/api/v1/routes/question_packs.py |
| PATCH  | /api/v1/question-packs/{pack_id}               | server/app/api/v1/routes/question_packs.py |
| POST   | /api/v1/question-packs/{pack_id}/questions     | server/app/api/v1/routes/question_packs.py |



## roadmaps

| Method | Path                                   | Source                               |
|--------|----------------------------------------|--------------------------------------|
| POST   | /api/v1/roadmaps                       | server/app/api/v1/routes/roadmaps.py |
| GET    | /api/v1/roadmaps/latest                | server/app/api/v1/routes/roadmaps.py |
| GET    | /api/v1/roadmaps/{roadmap_id}          | server/app/api/v1/routes/roadmaps.py |
| PATCH  | /api/v1/roadmaps/{roadmap_id}/progress | server/app/api/v1/routes/roadmaps.py |

## skills

| Method | Path           | Source                             |
|--------|----------------|------------------------------------|
| GET    | /api/v1/skills | server/app/api/v1/routes/skills.py |
| POST   | /api/v1/skills | server/app/api/v1/routes/skills.py |

## tests

| Method | Path                    | Source                            |
|--------|-------------------------|-----------------------------------|
| GET    | /api/v1/tests           | server/app/api/v1/routes/tests.py |
| POST   | /api/v1/tests           | server/app/api/v1/routes/tests.py |
| DELETE | /api/v1/tests/{test_id} | server/app/api/v1/routes/tests.py |
| GET    | /api/v1/tests/{test_id} | server/app/api/v1/routes/tests.py |
| PATCH  | /api/v1/tests/{test_id} | server/app/api/v1/routes/tests.py |

## users

| Method | Path                    | Source                            |
|--------|-------------------------|-----------------------------------|
| GET    | /api/v1/users           | server/app/api/v1/routes/users.py |
| GET    | /api/v1/users/{user_id} | server/app/api/v1/routes/users.py |
| PATCH  | /api/v1/users/{user_id} | server/app/api/v1/routes/users.py |